

The Language of Medicine

13th edition

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Chapter 20

Radiology and Nuclear Medicine

Chapter Goals (Slide 1 of 2)

- List the physical properties of x-rays.
- Identify diagnostic techniques used by radiologists and nuclear physicians.
- Name x-ray views and patient positions used in x-ray examinations.
- Describe the role of radioactivity in the diagnosis of disease.

Chapter Goals (Slide 2 of 2)

- Recognize medical terms used in the specialties of radiology and nuclear medicine.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 20

Lesson 20.1

Introduction

- **Radiology:** medical specialty concerned with the study of x-rays and other technologies (such as ultrasound and magnetic resonance) to produce and interpret images of the human body for diagnosis of disease
- **X-rays:** invisible waves of energy
- **Nuclear medicine:** use of radioactive substances in the diagnosis of disease

Personnel in This Medical Field

- Radiologist
- Nuclear physician
- Radiologic technologists
 - Radiographers, nuclear medicine technologists, sonographers

Characteristics of X-rays

- Expose photographic plates
- Penetrate substances
- Invisible
- Travel in straight lines
- Scatter
- Ionization

Diagnostic Techniques (Slide 1 of 2)

X-Ray studies

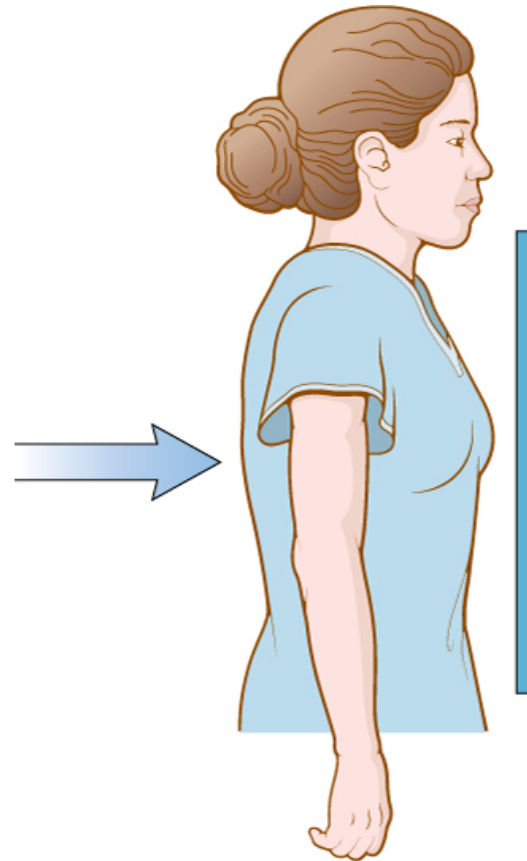
- Digital radiography
- Computed tomography (CT)
- Contrast studies
 - Barium sulfate: upper GI, lower GI
 - Iodine compounds: angiography, arthrography, cholangiography, digital subtraction angiography (DSA), hysterosalpingography, myelography, pyelography

Diagnostic Techniques (Slide 2 of 2)

- Fluoroscopy
- Digital imaging techniques
- Interventional radiology
- Ultrasound
- Magnetic imaging or magnetic resonance imaging (MRI or MR)

X-ray Positioning (Slide 1 of 6)

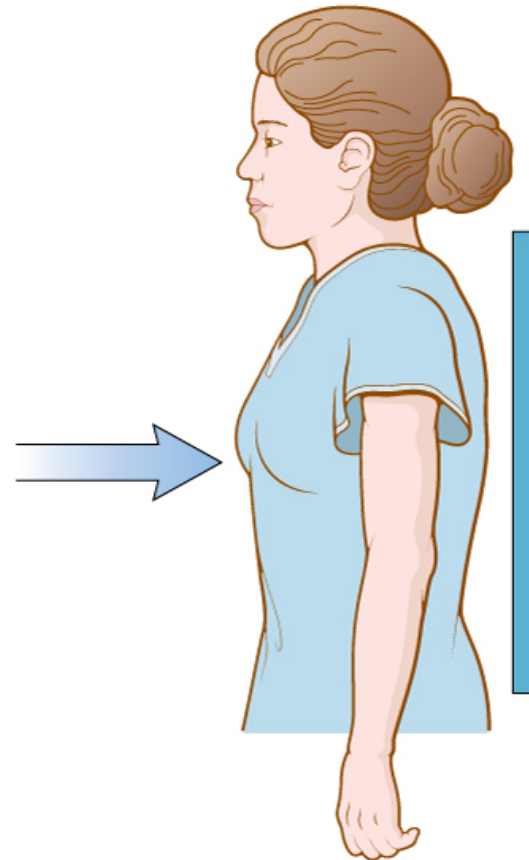
- Posteroanterior (PA) view: posterior source to anterior detector



Posteroanterior (PA) view

X-ray Positioning (Slide 2 of 6)

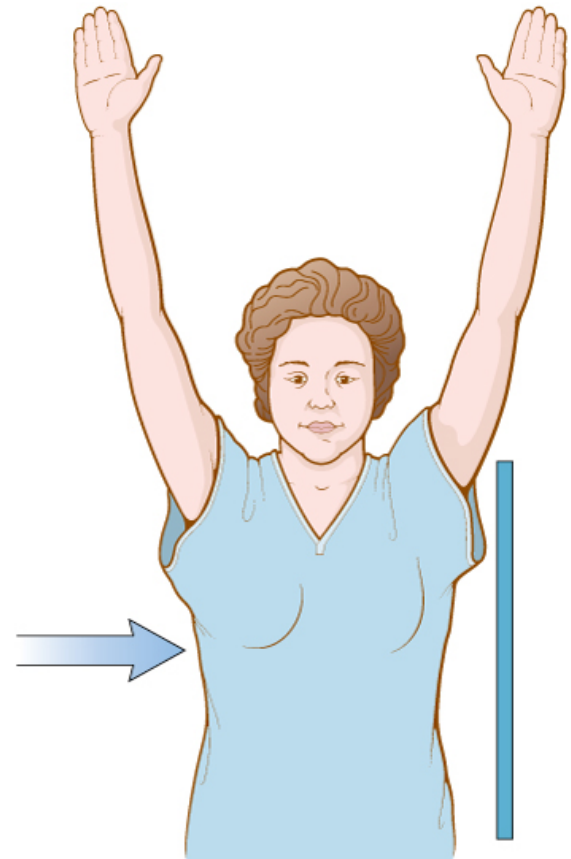
- Anteroposterior (AP) view: anterior source to posterior detector



Anteroposterior (AP) view

X-ray Positioning (Slide 3 of 6)

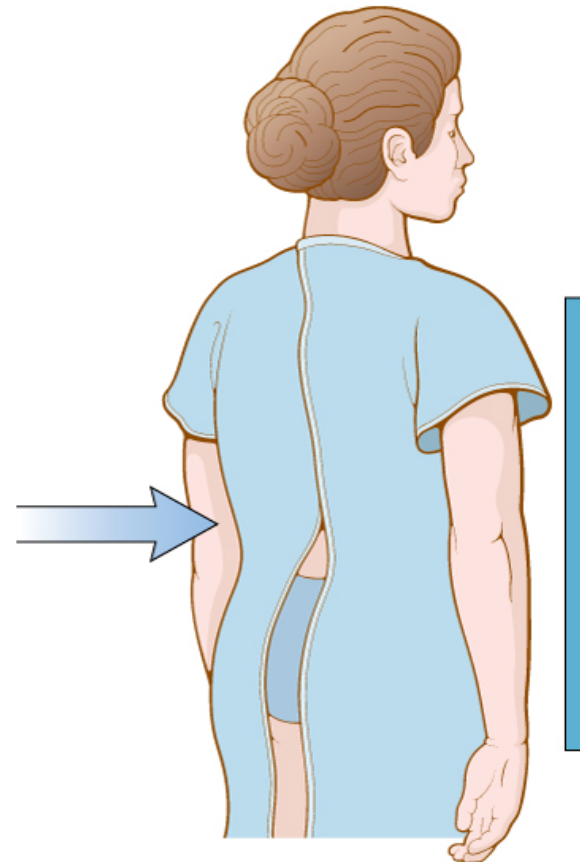
- **Lateral view:** in left lateral view, source at right of patient, to detector at left of patient



Left lateral view

X-ray Positioning (Slide 4 of 6)

Oblique view: source
slanting direction at
angle from
perpendicular plane



Oblique view

X-ray Positioning (Slide 5 of 6)

To describe the position of patient:

- Abduction—movement away from the midline
- Adduction—movement toward the midline
- Eversion—turning outward
- Extension—lengthening or straightening a flexed limb
- Flexion—bending a part of the body

X-ray Positioning (Slide 6 of 6)

To describe the position of patient:

- Decubitus—lying down on the side
- Prone—lying on the belly (face down)
- Recumbent—lying down (prone or supine)
- Supine—lying on the back (face up)

QUICK QUIZ (Slide 1 of 4)

**1. Obstructing the passage of
x-rays:**

- A. radiopaque
- B. radiolucent
- C. radioisotope
- D. radiopharmaceutical

Chapter 20

Lesson 20.2

Nuclear Medicine

- Radionuclides/radioisotopes
 - Alpha particles, beta particles, gamma rays
- Nuclear medicine tests
 - In vitro: test tube
 - In vivo: in the body

In Vitro Procedures

- Analysis of blood and urine
 - Radioimmunoassay (RIA) uses radioactive chemicals and antibodies to detect hormones and drugs in the patient's blood (for example, digitalis detection, hypothyroidism in newborns).

In Vivo Procedures

- Radioactive substance given directly to patient to evaluate organ function or image
 - Radiopharmaceutical (labeled compound) concentrates in organ
 - Scintiscanner (gamma camera) detection instrument produces picture (scintiscan)

Procedures Using Radionuclides

- Bone scan
- Lymphoscintigraphy
- Positron emission tomography (PET) scan
- PET/CT scan
- Single-photon emission computed tomography (SPECT)
- Technetium Tc-99m sestamibi (Cardiolite) scan
- Thallium (Tl) scan
- Thyroid scan

Focus on PET Scan

- Radioisotopes (emission of positrons) instead of contrast x-rays
- Intravenous injection
- Concentrates radioisotopes in tissues where the radionuclide is or is not being metabolized
- Useful in treating stroke, epilepsy, Alzheimer disease, brain tumors, abdominal and pulmonary malignancies

Focus on SPECT

- Intravenous injection of radioactive tracer
- Computer reconstruction of 3-D image based on many views
- Detects liver tumors, cardiac ischemia, diseases of bone and spine

Vocabulary (Slide 1 of 8)

Term	Meaning/Definition
computed tomography (CT)	Diagnostic x-ray procedure that produces a cross-sectional image of a specific body segment
contrast studies	Radiopaque materials are injected to obtain contrast with surrounding tissue when shown on x-ray film
gamma camera	Machine to detect gamma rays emitted from radiopharmaceuticals
gamma rays	High-energy rays emitted by radioactive substances in tracer studies

Vocabulary (Slide 2 of 8)

Term	Meaning/Definition
half-life	Time required for a radioactive substance to lose half its radioactivity by disintegration
interventional radiology	Therapeutic or diagnostic procedures performed by a radiologist
in vitro	Process, test, or procedure performed, measured, or observed <i>outside</i> a living organism
in vivo	Process, test, or procedure is performed, measured, or observed <i>within</i> a living organism

Vocabulary (Slide 3 of 8)

Term	Meaning/Definition
ionization	Transformation of electrically neutral substances into electrically charged particles
magnetic resonance imaging (MRI)	Use of magnetic field and radio waves to produce sagittal, coronal, and axial images of the body
nuclear medicine	Medical specialty that studies uses of radioactive substances in diagnosis of disease

Vocabulary (Slide 4 of 8)

Term	Meaning/Definition
positron emission tomography (PET)	Use of positron-emitting radioactive substances given intravenously to create a cross-sectional image of cellular metabolism
radioimmunoassay	Test that combines radioactive chemicals and antibodies to detect minute quantities of substances in a patient's blood
radioisotope	Radioactive form of an element ; radionuclide

Vocabulary (Slide 5 of 8)

Term	Meaning/Definition
radiolabeled compound	Radiopharmaceutical; used in nuclear medicine studies
radiology	Medical specialty concerned with the study of x-rays and their use in diagnosis of disease
radiolucent	Permitting the passage of x-rays
radionuclide	Radioactive form of an element; radioisotope

Vocabulary (Slide 6 of 8)

Term	Meaning/Definition
radiopaque	Obstructing the passage of x-rays
radiopharmaceutical	Radioactive drug used for diagnostic and therapeutic purposes
scan	Image of an area, organ or tissue of the body obtained from ultrasound, radioactive tracer studies, CT, or MRI
scintigraphy	Diagnostic nuclear medicine test using radiopharmaceuticals and gamma cameras to create images

Vocabulary (Slide 7 of 8)

Term	Meaning/Definition
single photon emission computed tomography (SPECT)	Radioactive tracer is injected intravenously and a computer reconstructs a 3D-image based on a composite of many views
tagging	Attaching a radionuclide to a chemical and following its path in the body
tracer studies	Use of radionuclide tags attached to chemicals and followed as they travel through the body

Vocabulary (Slide 8 of 8)

Term	Meaning/Definition
ultrasonography (US, U/S)	Diagnostic technique that projects and retrieves high-frequency sound waves as they echo off body parts
ultrasound transducer	Handheld device that sends and receives ultrasound signals
uptake	Rate of absorption of a radionuclide into an organ or tissue
ventilation-perfusion studies	Ventilation and perfusion of a radiopharmaceutical followed by imaging its passage through the respiratory tract

QUICK QUIZ (Slide 2 of 4)

2. Which of the following is a handheld device that sends and receives ultrasound signals?

- A. Gamma camera
- B. Roentgenology
- C. Radioisotope
- D. Transducer

Combining Forms, Suffixes, Prefixes, and Terminology (Slide 1 of 4)

Combining Form	Meaning
fluor/o	luminous
is/o	same
pharmaceut/o	drug
radi/o	x-rays
roentgen/o	x-rays

Combining Forms, Suffixes, Prefixes, and Terminology (Slide 2 of 4)

Combining Form	Meaning
son/o	sound
therapeut/o	treatment
vitro/o	glass
vivo/o	life

Combining Forms, Suffixes, Prefixes, and Terminology (Slide 3 of 4)

Suffix	Meaning
-gram	record
-graphy	process of recording
-lucent	to shine
-opaque	obscure

Combining Forms, Suffixes, Prefixes, and Terminology (Slide 4 of 4)

Prefix	Meaning
cine-	movement
echo-	a repeated sound
ultra-	beyond

Abbreviations (Slide 1 of 4)

Abbreviation	Meaning
Angio	Angiography
AP	Anteroposterior
Ba	Barium
BE	Barium enema
C-spine	Cervical spine films
CT	Computed tomography
CXR	Chest x-ray (film)
Decub	Decubitus – lying down
DICOM	Digital image communication in medicine
DI	Diagnostic imaging

Abbreviations (Slide 2 of 4)

Abbreviation	Meaning
DSA	Digital subtraction angiography
ECHO	Echocardiography
EUS	Endoscopic ultrasonography
^{18}F -FDG	Fluorodeoxyglucose
Gd	Gadolinium
^{123}I	Isotope of radioactive iodine
^{131}I	Isotope of radioactive iodine
IVP	Intravenous pyelogram
KUB	Kidneys, ureters, bladder (series)
LAT	Lateral

Abbreviations (Slide 3 of 4)

Abbreviation	Meaning
LS films	Lumbosacral (spine) films
L-spine	Lumbar spine
MDCT	Multidetector CT scanner
MR, MRI	Magnetic resonance, magnetic resonance imaging
MRA	Magnetic resonance angiography
MRV	Magnetic resonance venography
MUGA	Multiple-gated acquisitions (scan)
PA	Posteroanterior
PACS	Picture archival and communications system
PET	Positron emission tomography
PET/CT	Positron emission tomography/computed tomography
RAIU	Radioactive iodine uptake (test)

Abbreviations (Slide 4 of 4)

Abbreviation	Meaning
RFA	Radiofrequency ablation
SBFT	Small bowel follow-through
SPECT	Singe photon emission computed tomography
⁹⁹ Tc	Radioactive technetium
²⁰¹ Tl	Thallium-201
T-spine	Thoracic spine
UGI	Upper gastrointestinal (series)
US, U/S	Ultrasound; ultrasonography
V/Q scan	Ventilation-perfusion scan of the lungs

QUICK QUIZ: (Slide 3 of 4)

3. Which term refers to movement away from the midline of the body?

A. Adduction

B. Abduction

C. PA

D. Inversion

QUICK QUIZ (Slide 4 of 4)

4. Which term refers to a patient lying on the back?

- A. Prone
- B. Supine
- C. Flexion
- D. Decubitus